



UPC INSTRUMENTS PRIVATE LIMITED

***Cleaning & Maintenance Procedure for
Online Water Analyzer***

Model: 'UPC-WA-202'

Parameters: BOD, COD, TSS, pH

1. OBJECTIVE

To ensure long-term accuracy, reliability, and operational efficiency of the Online Water Analyzer by performing regular automated and manual cleaning, along with preventive maintenance.

2. CLEANING SYSTEM OVERVIEW

- The analyzer is equipped with an **automated internal cleaning mechanism**.
 - **H₂SO₄ dosing pump (5% concentration)** is installed for chemical cleaning of the flow cell and sensor chambers.
 - Cleaning is **controlled via relay logic**, executed automatically every **24 hours**.
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3. AUTOMATIC CLEANING PROCEDURE

Frequency: Once every 24 hours (configurable)

Process Flow:

1. **Triggering:**
Relay is activated automatically via the system clock/timer at predefined intervals (every 24 hours).
2. **Dosing Cycle:**
 - The **5% H₂SO₄** solution is pumped into the flow chamber.
 - Contact time is maintained to dissolve organic/inorganic deposits from the UV-Vis and pH sensors.
 - Residual acid is flushed using distilled/sample water through the sampling pump.
3. **Completion:**
 - Relay turns OFF the dosing pump after the predefined cleaning cycle.
 - System resumes normal sampling and analysis.

 *Do not interrupt the system during the cleaning cycle unless in emergency mode.*

4. MANUAL CLEANING PROCEDURE (If Required)

Frequency:

- Yearly
- When automated cleaning is insufficient or system shows fouling alarms

Steps:

1. **Deactivate analyzer** from UI before manual intervention.

2. Open the sensor/flow chamber carefully.
3. Remove and inspect the sensors (UV-Vis and pH electrode).
4. **Clean flow cell and sensors** with mild non-abrasive brush and H_2SO_4 ($\leq 5\%$) or manufacturer-recommended cleaning solution.
5. Rinse thoroughly with **distilled water**.
6. Reinstall components and restart the system.
7. Observe 2–3 stable cycles to ensure performance.
8. Do not perform manual cleaning without taking support with our helpline numbers

5. PERIODIC MAINTENANCE CHECKLIST

Interval Task

Daily Ensure cleaning cycle is executed; check H_2SO_4 level in dosing tank

Weekly Inspect sampling pump suction and discharge lines

Monthly Manually clean sensors if heavy fouling is observed. Clean it by passing 10% H_2SO_4

Quarterly Verify calibration status and recalibrate if drift is observed

Annually Replace pH probe if deviation persists despite recalibration. Otherwise not required.

6. PRECAUTIONS & SAFETY MEASURES

- Always wear **PPE (gloves, goggles, lab coat)** while handling H_2SO_4 .
- **Label and secure the H_2SO_4 tank** with appropriate hazard symbols.
- Ensure **good ventilation** around the analyzer cabinet.
- Use **only recommended concentrations ($\leq 5\%$)** for chemical dosing.
- Disconnect power during **manual internal cleaning** or servicing.

7. TROUBLESHOOTING INDICATIONS

Symptom	Action
Cloudy or unstable readings	Initiate manual cleaning cycle
No auto-cleaning observed	Check relay, wiring, and cleaning timer configuration
Acid not dispensing	Check dosing pump power, tube blockages, or relay

Symptom	Action
pH drift persists	Replace pH probe or clean electrode thoroughly

8. LOGGING & DOCUMENTATION

All cleaning and maintenance activities must be recorded in the **Analyzer Logbook** with:

- Date & Time
- Performed By
- Type of Cleaning (Auto/Manual)
- Observations (e.g., fouling, calibration drift)
- Corrective Action Taken
- Signature

9. TECHNICAL SUPPORT

For any issues beyond routine maintenance, contact:

Email: info@unitechmeter.com

Technical Support: +91 70822 00823

Website: www.unitechmeter.com

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